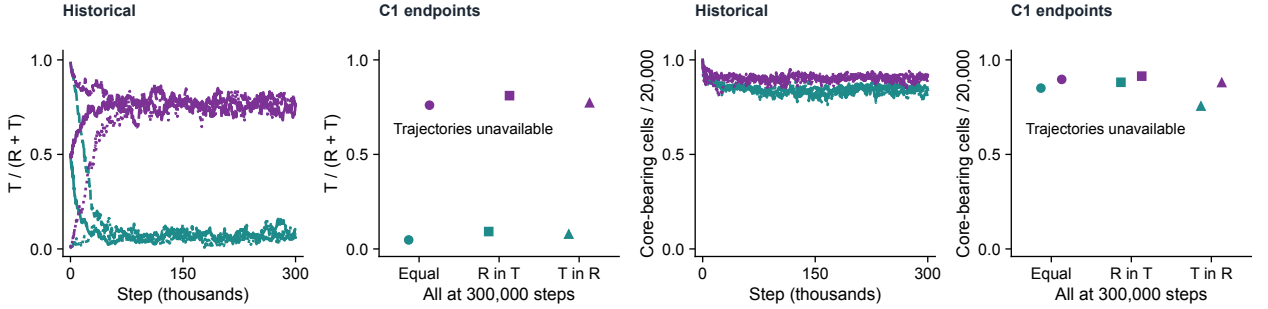
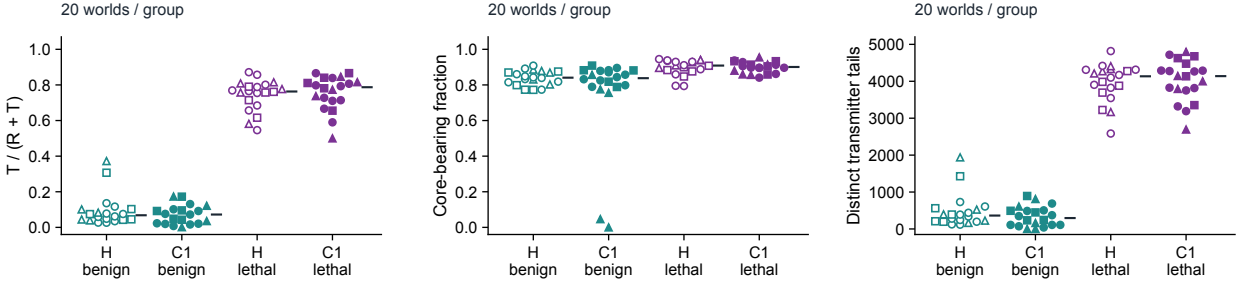
a **Matched chemistry, three starting mixtures**



Benign: teal; lethal: purple. Equal: solid/circle; R invades: dash/square; T invades: dot/triangle.

L32, mutation on; medians of 10 equal-start or 5 invasion worlds/chemistry/cohort. C1 offsets separate chemistries.

b **Endpoints keep every world**

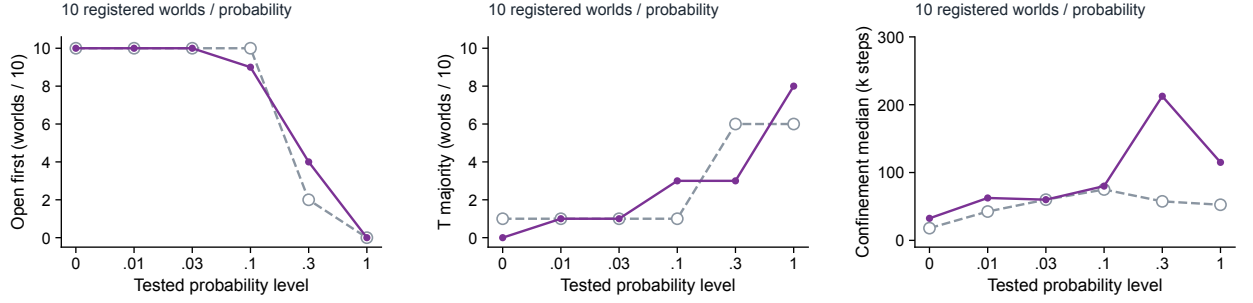


Circle: equal (10); square: R invades T (5); triangle: T invades R (5).

Missing ratios: × 1 C1 benign core-loss world, excluded from T median; retained in abundance/tails.

Open: historical; filled: C1. Black bars at right: medians of defined values; horizontal offsets avoid collisions.

c **Lethality dial: historical and C1**



DZ1: historical PASS; C1 PASS

DZ3: historical FAILED component; C1 PASS

DZ2: historical FAILED component; C1 PASS

Grey open/dashed: historical; purple: C1. KM: all 10 worlds, nondetections censored at 300k; p=.3 → 1 timing reversal.